

The Only Child

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Motivation

There are now more only children worldwide than ever before:

- ▶ 22% of all U.S. families (Pew Research Center, 2015)
- ▶ 70% of all urban families in China (NBS China, 2007)
- ▶ 21% of Danish families (Statistics Denmark, 2020)

Would only children do better or worse if they had siblings?

- ▶ **May do better** — they do not share family resources (Becker and Lewis, 1973; Becker and Tomes, 1976)
- ▶ **May do worse** — they lack younger siblings to teach or socialize with (Zajonc and Markus, 1975; Zajonc, 1976)

Quantifying the effect of siblings is difficult due to selection:

- ▶ Families with one child may differ in values, stability, and other unobservables.

This Paper

- ▶ What is the effect of siblings on firstborns?
 - ▶ Would the cognitive and non-cognitive development of firstborn children improve or worsen if they had siblings?
- ▶ Natural experiment: in-vitro fertilization (IVF).
 - ▶ Focus on families who underwent the procedure for a second child, comparing firstborns in families whose procedure succeeded versus failed.
- ▶ Estimate effects on school performance, personality, and well-being using a near-universe of Danish firstborns whose parents underwent IVF.
 - ▶ Math, language, conscientiousness, agreeableness, emotional stability, and happiness.

Literature and Contribution

1. Studies exploiting China's one-child policy to estimate the effect of having siblings

- ▶ H. Li et al. (2008); Rosenzweig & Zhang (2009); Qian (2009); Cameron et al. (2013); Liu (2014); B. Li & Zhang (2017); Guo et al. (2020); Xiao (2024)

We provide the first evidence from a developed country, where resource constraints play a smaller role and the socioeconomic context is held fixed.

2. Studies on additional siblings (≥ 2), leveraging twin births and sex composition

- ▶ Conley & Glauber (2006); Black et al. (2005); Åslund & Grönqvist (2010); Angrist et al. (2010); De Haan (2010)

We provide evidence on the first-sibling margin, where both resource dilution and social-learning effects are likely strongest.

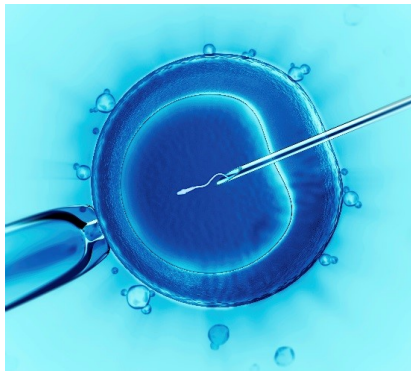
3. Birth order literature comparing earlier-born to later-born children

- ▶ Black et al. (2011); Conley & Glauber (2006); Gary-Bobo et al. (2006); Kantarevic & Mechoulam (2006); Kristensen & Bjerkedal (2007); Booth & Kee (2009); De Haan (2010); Houmark (2023)

We shed light on the “only-child handicap” (Zajonc & Markus, 1975).

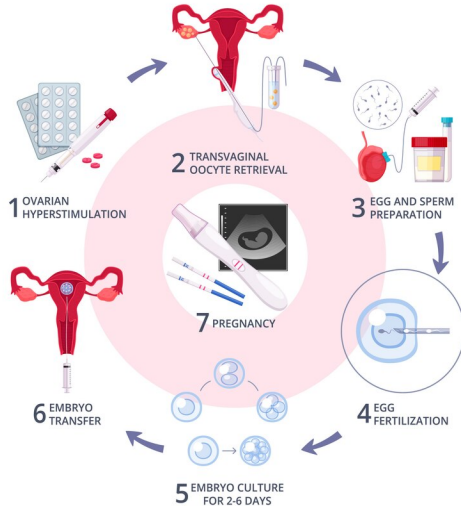
In Vitro Fertilisation

- ▶ Couples who have trouble conceiving can attempt IVF.
- ▶ Treatment success has a random component.
 - ▶ Uncorrelated with firstborn and parent observables conditional on mother's age.
- ▶ Natural experiment:
 - ▶ First children whose parents have a successful treatment get a sibling.
 - ▶ First children whose parents have an unsuccessful treatment are more likely to remain the only child.



Source: CDC

In Vitro Fertilisation



Source: Freepik

Data and Sample

- ▶ Danish IVF register: treatment dates and outcomes.
- ▶ Administrative registers linking parents and children.
- ▶ **Cognitive outcomes:** nationwide reading tests (grades 2, 4, 6, and 8) and math tests (grades 3 and 6).
- ▶ **Noncognitive outcomes:** nationwide school surveys measuring wellbeing and three Big Five personality traits:
 - ▶ Conscientiousness, agreeableness, emotional stability
 - ▶ Openness to experience, extraversion
- ▶ Outcomes are standardized by cohort and averaged within subject or trait.
- ▶ Main sample: first-born children of mothers seeking IVF for a second child:
 - ▶ **11,000** with cognitive outcomes; **7,500** with noncognitive outcomes

Identification

Treatment:

$$S = \begin{cases} 1 & \text{if the first child has a sibling,} \\ 0 & \text{if the first child remains an only child.} \end{cases}$$

Instrument:

$$Z = \begin{cases} 1 & \text{if the first IVF attempt succeeds,} \\ 0 & \text{if the first IVF attempt fails.} \end{cases}$$

Compliers: children who have a sibling only if the first IVF attempt succeeds.

Assumptions

1. IVF success is as good as random (conditional on mother's age).
2. IVF success affects outcomes only through having a sibling.

$$\frac{E[Y|Z=1] - E[Y|Z=0]}{E[S|Z=1] - E[S|Z=0]} = E[Y(1) - Y(0) \mid C=1].$$

Raw Relationship Between Having Siblings and Child Outcomes

Table 1: Associations between having siblings and school outcomes

	Math Test	Reading Test	Agreeable	Consc.	Emotional Stability	School Happiness
<i>Panel A: Representative sample</i>						
Having siblings	0.171 (0.005)	0.090 (0.004)	0.074 (0.005)	0.122 (0.005)	0.114 (0.005)	0.115 (0.005)
Observations	256,992	339,281	224,486	224,571	224,537	224,522
<i>Panel B: IVF sample</i>						
Having siblings	0.088 (0.023)	0.054 (0.020)	0.030 (0.021)	0.051 (0.022)	0.042 (0.022)	0.074 (0.021)
Observations	8,308	10,906	7,689	7,666	7,567	7,783

Note: Unconditional outcome differences between first-born children with and without siblings. Standard errors in parentheses.

Causal Estimates

Table 2: First-stage and second-stage estimates

	Math Test Sample	Reading Test Sample	Agreeable Sample	Consc. Sample	Emotional Stability Sample	School Happiness Sample
<i>Panel A: First-stage</i>						
Success	0.344 (0.005)	0.338 (0.005)	0.324 (0.006)	0.323 (0.006)	0.323 (0.006)	0.324 (0.006)
Observations	8,308	10,906	7,689	7,666	7,567	7,783
<i>Panel B: Second-stage</i>						
Having siblings	-0.054 (0.069)	-0.049 (0.059)	-0.038 (0.068)	-0.025 (0.071)	-0.075 (0.071)	-0.118 (0.068)
Observations	8,308	10,906	7,689	7,666	7,567	7,783

Note: Panel A—first-stage estimates of the effect of parents' first IVF attempt succeeding on the likelihood that the first-born has at least one sibling (complier share). Panel B—IPW estimates of the effect of having siblings on standardized outcomes. Standard errors in parentheses.

Improving Identification

Our causal estimates rule out the OLS estimates, though some limitations remain.

- ▶ Estimates are relatively imprecise.
- ▶ Identification covers only compliers.
- ▶ Potential exclusion violations may arise if age gaps between siblings matter.

To address this, we instead exploit variation from the *entire IVF sequence*.

- ▶ Treatment group: families whose first IVF attempt succeeds.
- ▶ New control group: families for whom all IVF attempts fail.

Key issue: the new control group is a selected sample:

- ▶ Families with more IVF attempts are underrepresented.

Key idea:

- ▶ Among families whose IVF sequence fails, we observe the number of attempts.
- ▶ Using per-attempt success rates, we can infer how likely they were to fail ex-ante.
- ▶ We reweight observations to construct a representative control group.

Estimates cover families dependent on the entire IVF sequence and are more precise.

Improved Causal Estimates

Table 3: Improved first-stage and second-stage estimates

	Math Test Sample	Reading Test Sample	Agreeable Sample	Consc. Sample	Emotional Stability Sample	School Happiness Sample
<i>Panel A: First-stage</i>						
Success	0.555 (0.008)	0.546 (0.008)	0.537 (0.010)	0.538 (0.010)	0.537 (0.011)	0.538 (0.010)
Observations	5,961	7,783	5,379	5,350	5,283	5,446
<i>Panel B: Second-stage</i>						
Having siblings	-0.024 (0.045)	-0.006 (0.035)	-0.007 (0.045)	-0.008 (0.047)	-0.016 (0.049)	-0.031 (0.042)
Observations	5,961	7,783	5,379	5,350	5,283	5,446

Note: Panel A reports first-stage estimates of the effect of parents' IVF attempts succeeding on the likelihood that the first-born has at least one sibling (relier share). Panel B reports the second-stage estimates of the effect of having siblings on standardized outcomes for reliefs. Standard errors in parentheses.

Summary

- ▶ We find little evidence that siblings affect personality traits or school performance.
- ▶ All estimated effects are small and statistically insignificant.
- ▶ When examining heterogeneity in the only-child effect, the estimates remain remarkably stable across family income, fertility-related health conditions, maternal age, and the child's age and gender.
- ▶ Observed differences between only children and others likely reflect selection rather than causal effects of having siblings.

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Balance

	Success Mean (1)	Failure Difference (2)	Conditional Difference (3)
Female child	0.49	0.01	0.01
IVF child	0.50	0.03	0.01
Child weight (kg)	3.52	0.01	0.01
Child length (cm)	52.05	-0.00	0.03
Mother age	33.41	-1.12	-0.03
Mother college	0.52	-0.00	0.02
Mother income	0.21	-0.00	0.01
Mother work	0.93	0.01	0.01
Father age	35.87	-1.07	0.00
Father college	0.37	-0.01	0.01
Father income	0.27	-0.00	0.01
Father work	0.91	0.00	-0.00
Joint p -value		0.00	0.38

Note: Observations=10,906. Success probability=0.30. "Failure Difference" compares families whose first IVF attempt succeeded versus failed. "Conditional Difference" adjusts for parents' age (and squares) and treatment year indicators using inverse probability weights.

Model

Families differ in two unobserved characteristics:

- ▶ “Willingness” to undergo IVF, $W \in \{1, \dots, \bar{w}\}$
 - ▶ Would undergo W IVF procedures for the second child if all previous attempts failed.
- ▶ “Reliance” on IVF, $R \in \{0, 1\}$
 - ▶ Would have no second child if all IVF attempts failed ($R = 1$).

Observables:

- ▶ IVF attempt j success indicator: Z_j
- ▶ Number of realized attempts: $A = \min(\{j : Z_j = 1\} \cup \{W\})$
- ▶ Second-child indicator: $S = \max(Z_A, 1 - R)$

Assumption:

A3. Sequential Independence: $Y_1(1), Y_0(0), Y_0(1), W, R \perp Z_j \mid A > j$.

- ▶ Among those who attempt IVF j times, j th attempt success is as good as random.

Simple World: Max 2 Procedures, All Reliers

[Back](#)

$$W = 1$$

(willing to try once)

$$Z_1 = 1$$

$$Z_1 = 0$$

Simple World: Max 2 Procedures, All Reliers

[Back](#)

$$W = 1$$

(willing to try once)

$$Z_1 = 1$$

$$Z_1 = 0$$

$$W = 2$$

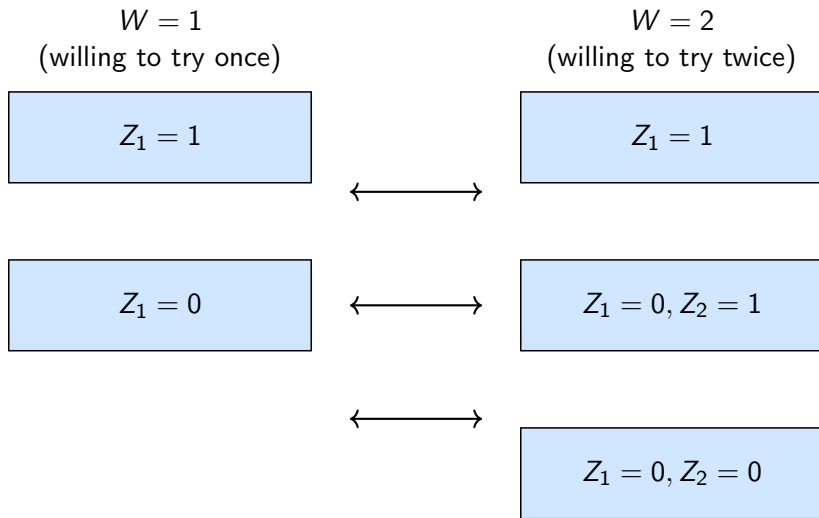
(willing to try twice)

$$Z_1 = 1$$

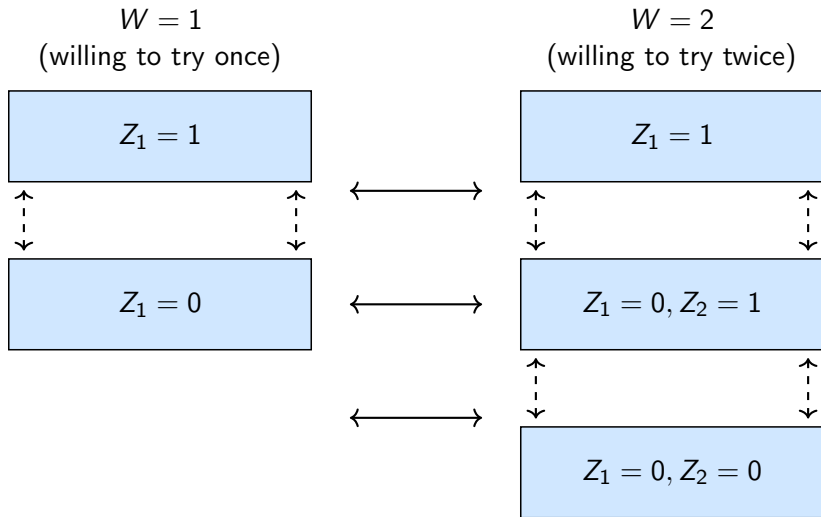
$$Z_1 = 0, Z_2 = 1$$

$$Z_1 = 0, Z_2 = 0$$

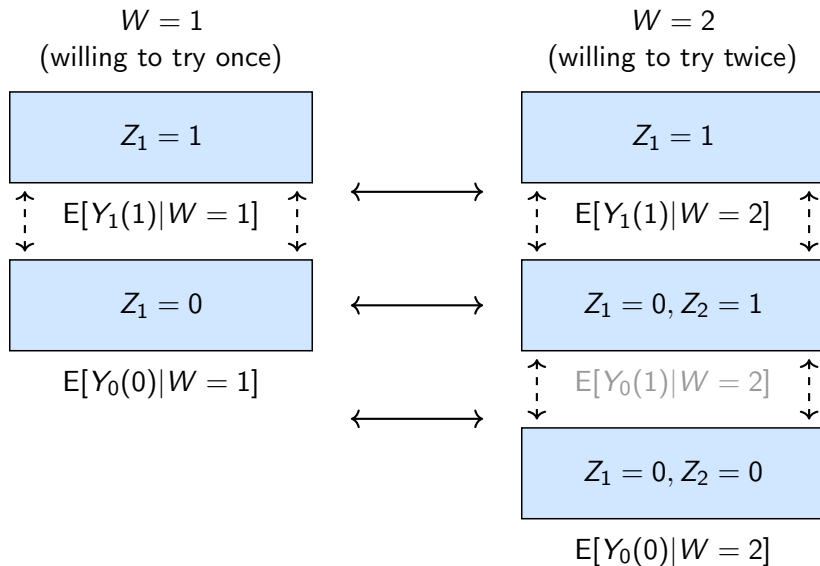
Simple World: Max 2 Procedures, All Reliers

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Simple World: Max 2 Procedures, All Reliers [Back](#)



Simple World: Max 2 Procedures, All Reliers [Back](#)



Simple World (Observed): Max 2 Procedures, All Reliers

[Back](#)

$$W = 1$$

(willing to try once)

$$W = 2$$

(willing to try twice)

$$Z_1 = 1$$

$$Z_1 = 0$$

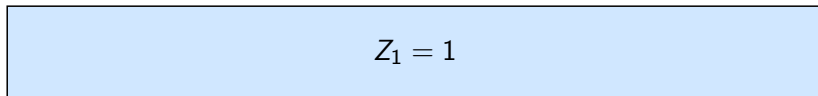
$$Z_1 = 0, Z_2 = 1$$

$$Z_1 = 0, Z_2 = 0$$

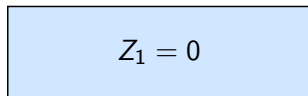
Simple World (Observed): Max 2 Procedures, All Reliers [Back](#)

$W = 1$
(willing to try once)

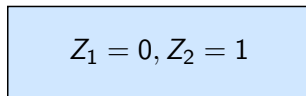
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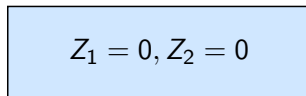
$E[Y_1(1)]$



$E[Y_0(0)|W = 1]$



$E[Y_0(1)|W = 2]$

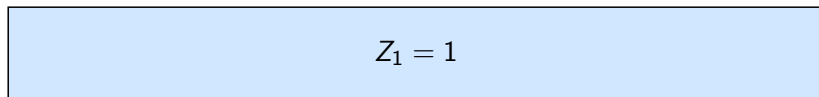


$E[Y_0(0)|W = 2]$

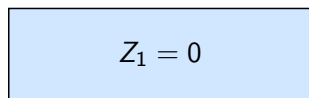
Simple World (Observed): Max 2 Procedures, All Reliers [Back](#)

$W = 1$
(willing to try once)

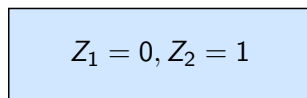
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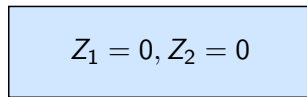
$E[Y_1(1)]$



$E[Y_0(0)|W = 1]$



$E[Y_0(1)|W = 2]$

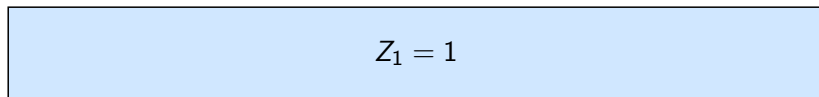


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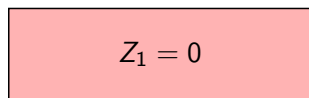
Simple World (Observed): Max 2 Procedures, All Reliers [Back](#)

$W = 1$
(willing to try once)

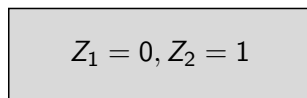
$W = 2$
(willing to try twice)



$E[Y_1(1)]$

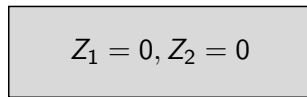


$E[Y_0(0)|W = 1]$



$E[Y_0(1)|W = 2]$

$$Pr(W = 1) = \frac{\text{red area}}{\text{red area} + \text{gray area}}$$



$E[Y_0(0)|W = 2]$